

Project Design - Proposed Solution

Parameter	Detail
Date	18 June 2026
Team ID	Book-a-Doctor Platform
Project Name	Book a Doctor
Maximum Marks	2 Marks

Proposed Solution Specification Table

S.No.	Parameter	Description / Specification for "Book a Doctor"
1	Problem Statement	- Patient difficulty in finding verified medical specialists. - Clinic coordination clashes and double-bookings. - Insecure file transfer of patient health histories.
2	Idea / Solution description	Book a Doctor is a MERN stack healthcare portal with: - Admin vetting dashboard for doctor credentials. - Mongoose validated scheduling engine. - Multer powered PDF document locker. - In-app notification dispatcher. - Digital prescriptions.
3	Novelty / Uniqueness	- Glassmorphism UI with native Light/Dark toggle. - Dual-token JWT (Access Token + HttpOnly Cookies Refresh Token) preventing XSS hijacks. - Role-Based Access Control protecting routes at both backend middleware and client-side levels.
4	Social Impact / Customer Satisfaction	- Empowers patients to verify doctor credentials before booking, reducing fraud. - Eliminates paper overhead and clinic waiting delays. - Securely keeps history files private to authorized consultations.
5	Business Model (Revenue Model)	- Commission Model: Small platform fee on successful bookings. - Premium Listing: Doctors pay to feature at the top of filter results. - SaaS Integration: Clinics pay monthly subscriptions to integrate their scheduling.
6	Scalability of the Solution	- Frontend build compiles to static assets (Vercel/Netlify CDNs). - Backend Express APIs can be deployed as serverless functions or containerized (Docker) to scale microservices. - Database hosted on MongoDB Atlas sharded clusters.